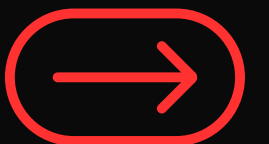


AI-Powered Mystery Case Analysis and Investigation Assistant



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PROBLEM STATEMENT

- Mystery investigations often involve large amounts of information.
- Investigators need to examine clues, suspects, statements and evidence carefully.
- Finding relevant information from multiple documents can be time-consuming.
- Manually connecting clues with available knowledge can lead to missed information.
- A smart AI assistant is needed to retrieve relevant investigation knowledge and provide a clear response.



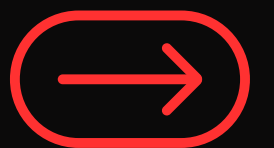


PROJECT OBJECTIVE

To build an AI-powered investigation assistant that uses Retrieval-Augmented Generation (RAG) to analyze user queries related to mystery cases.

The system should:

- Retrieve relevant information from the knowledge base.
- Use an LLM to analyze the retrieved information.
- Provide a clear and understandable answer.
- Help users investigate clues and case information efficiently.





EXISTING SYSTEM

01 — Manual Investigation

Investigators manually search through case documents and clues.

02 — Keyword Search

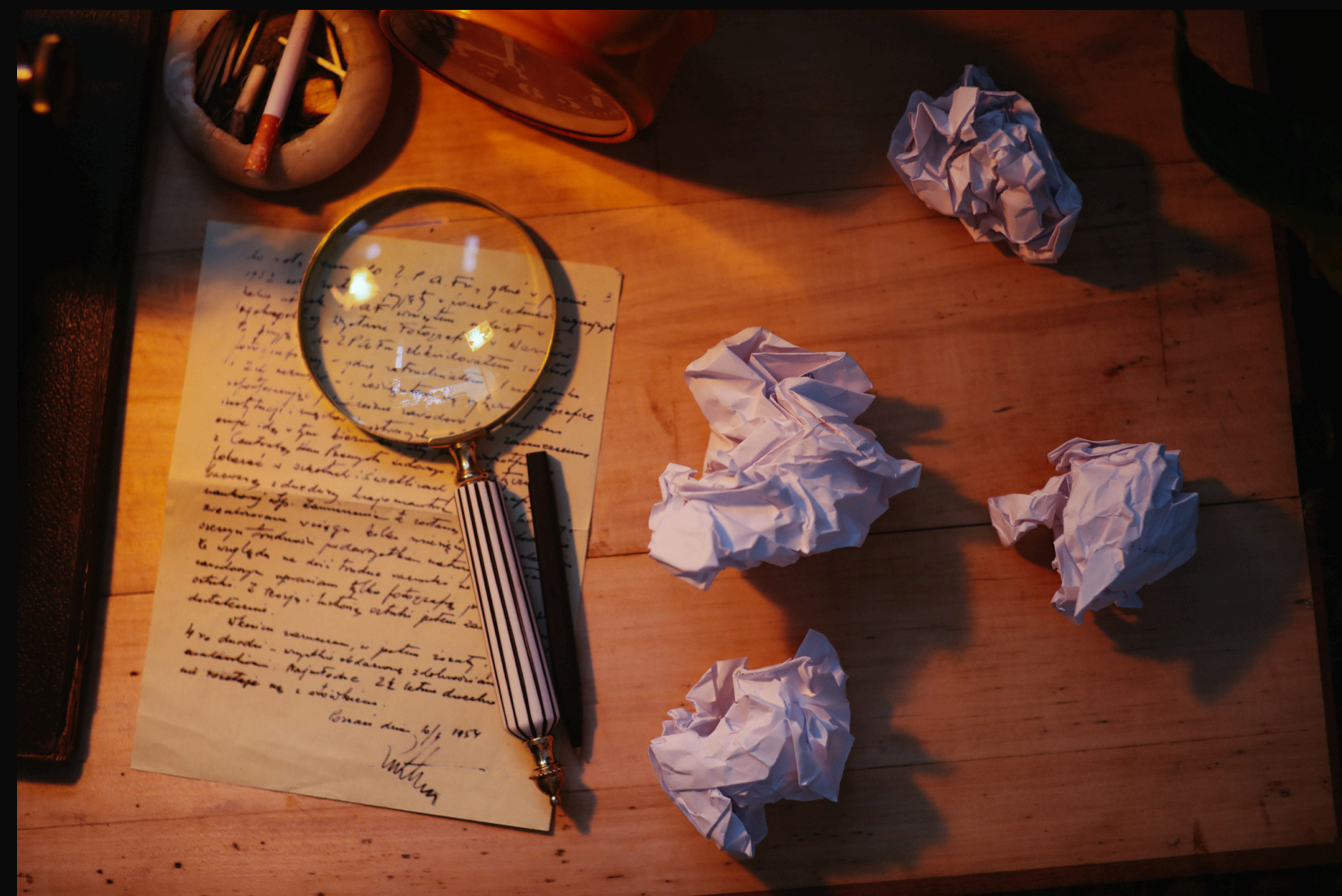
Traditional search systems mainly find information based on matching keywords.

03 — General AI Chatbots

General AI models may answer questions without using the specific investigation knowledge base.

Limitations

- Time-consuming
- Difficult to search large documents
- May miss relevant evidence
- No investigation-specific knowledge retrieval





Proposed Gen AI Solution

Mystery Investigation AI uses RAG (Retrieval-Augmented Generation).

The system first retrieves relevant investigation information from the Knowledge Base.

Then the retrieved information is provided to the AI model, which generates the final response.

RAG Flow:

User Query



Knowledge Retrieval 



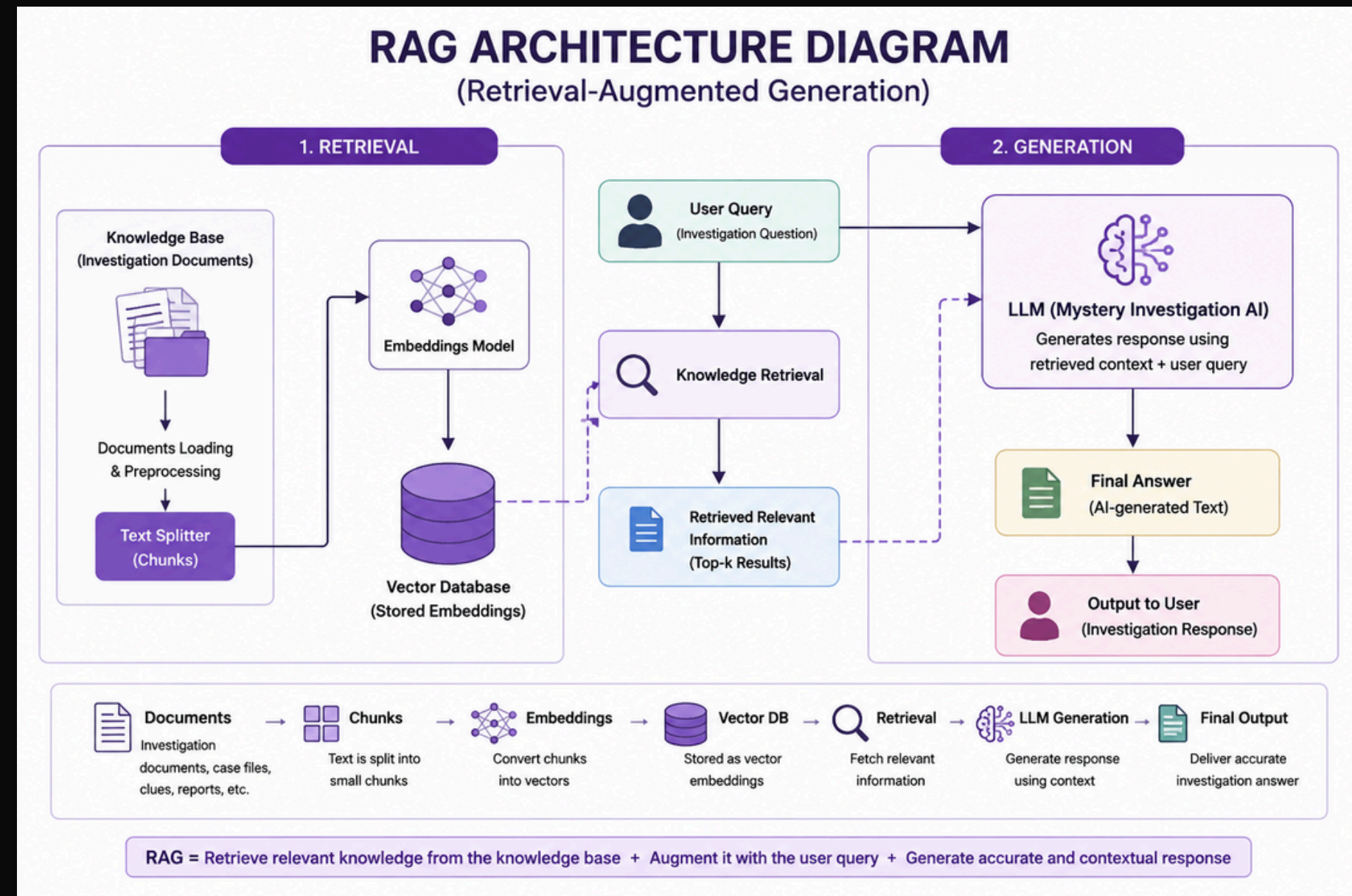
Relevant Investigation Knowledge



Mystery Investigation AI 



Final Answer



Tools & Technologies Used

Tools & Technologies

1. PLATFORM



Dify AI Platform

Used to build the AI workflow and application.

2. AI MODEL



LLM

Large Language Model used to understand the query and generate responses.

3. RETRIEVAL



Knowledge Retrieval

Retrieves relevant information from the knowledge base based on the user query.

4. KNOWLEDGE BASE



Investigation Documents

Collection of mystery investigation documents used as the knowledge base.

5. ARCHITECTURE



RAG

Retrieval-Augmented Generation architecture combines retrieval and generation for accurate answers.

6. OUTPUT



AI-generated Text

The final response generated by the AI based on retrieved information.

7. INTERFACE



AI Workflow

Workflow interface in Dify to connect components and build the application.



Workflow

Step 1

User enters a mystery investigation question.

Step 2

The question is sent to Knowledge Retrieval.

Step 3

Knowledge Retrieval searches the connected knowledge base.

Step 4

Relevant information is retrieved.

Step 5

Retrieved information is given to Mystery Investigation AI.

Step 6

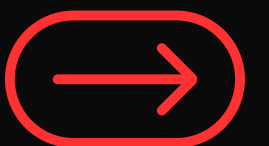
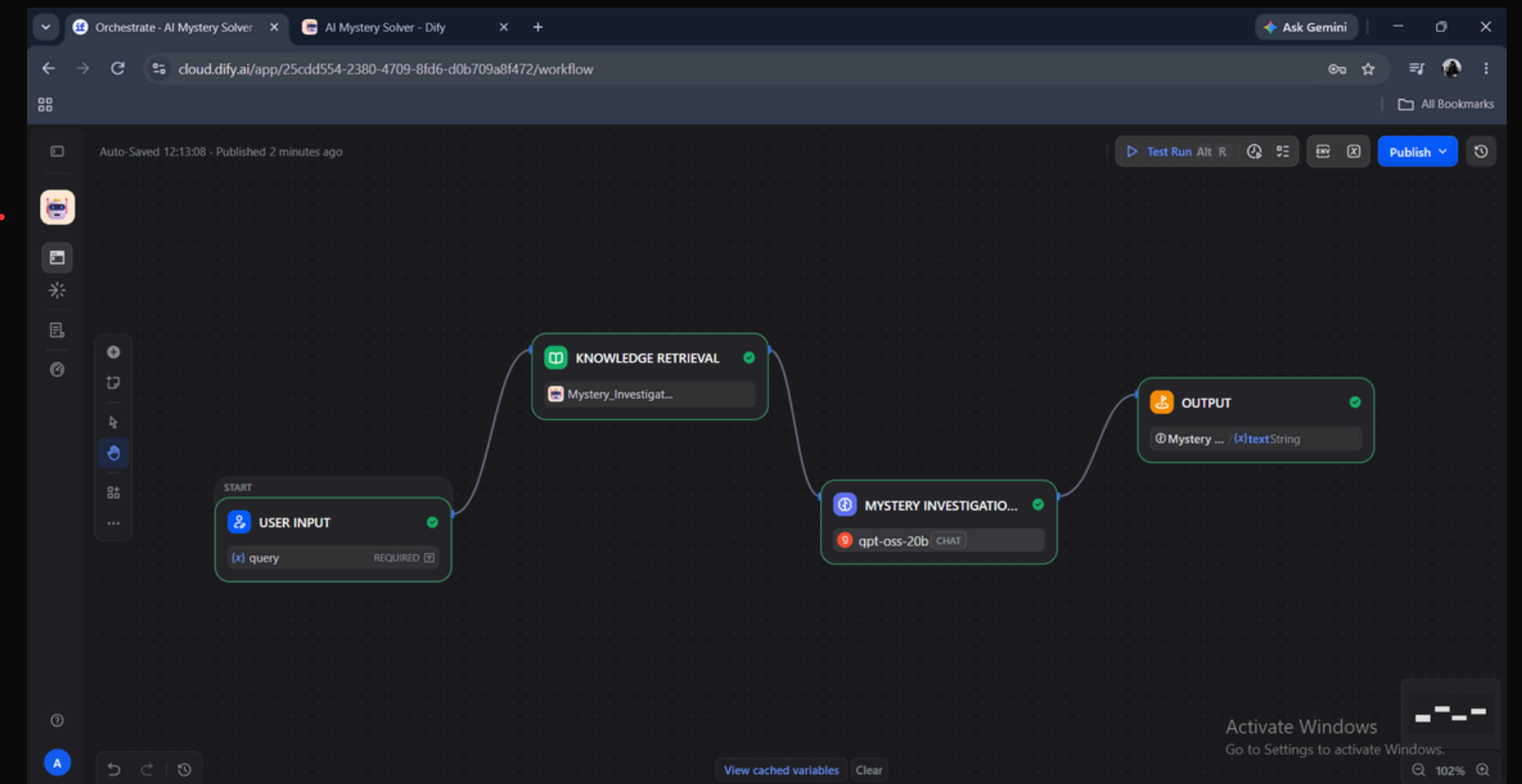
The AI analyzes the question with the retrieved context.

Step 7

AI generates the final answer.

Step 8

The final answer is displayed through the Output node.





WORK PICTURES

Orchestrate · AI Mystery Solver · ×

AI Mystery Solver - Dify · ×

AI Mystery Solver - Dify · ×

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 AI Mystery Solver

Run Once

Run Batch

query

"A diamond necklace is missing from a jewellery showroom. Three employees were prese

Clear

▶ Execute



AI will give you what you want here.

POWERED BY Dify

Activate Windows
Go to Settings to activate Windows.





WORK PICTURES



Orchestrate - AI Mystery Solver

AI Mystery Solver - Dify

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Ask Gemini

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AI Mystery Solver

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Run Once

Run Batch

query

"A diamond necklace is missing from a jewellery showroom. Three employees were prese

Clear

▶ Execute

Workflow Process

USER INPUT

KNOWLEDGE RETRIEVAL

MYSTERY INVESTIGATION AI

OUTPUT

RESULT

DETAIL

> Thought(0.0s)

Case Summary (Facts Only)

- A diamond necklace was reported missing from a jewellery showroom.
- Three employees were present at the showroom at the time of the incident.
- No other persons, objects, or events have been mentioned in the brief.

Orchestrate - AI Mystery Solver

AI Mystery Solver - Dify

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Run Once

Run Batch

query

"A diamond necklace is missing from a jewellery showroom. Three employees were prese

Clear

▶ Execute

1. Important Facts

#	Fact	Source
1	The item missing is a diamond necklace.	Case description
2	The incident occurred inside a jewellery showroom.	Case description
3	Three employees were present at the time of the incident.	Case description

2. Important Clues (As Provided)

#	Clue	Status
1	Presence of three employees	Incomplete – no details on their roles, actions, or alibis

3. Possible Suspects Requiring Further Investigation

#	Suspect	Why they're a suspect
1	Employee A	Had access to the showroom; no alibi given
2	Employee B	Same as above
3	Employee C	Same as above

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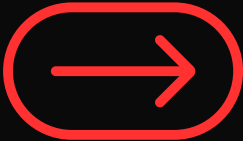
Dify

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Go to Settings to activate Windows.



WORK PICTURES



Orchestrate - AI Mystery Solver

AI Mystery Solver - Dify

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Ask Gemini

AI Mystery Solver

Run OnceRun Batch

query

*A diamond necklace is missing from a jewellery showroom. Three employees were prese

ClearExecute

4. Access & Opportunity

Access: All three employees had legitimate access to the showroom.

Opportunity: The necklace was in their shared work area; any of them could have taken it without immediate detection, especially if no real-time monitoring was in place.

5. Timeline (Based on Available Data)

Time	Event
T0	Showroom open for business
T1	Employees present (exact time unknown)
T2	Necklace missing (exact time unknown)

Missing: precise timestamps for opening, employee arrival, and discovery of the missing necklace.

6. Possible Motives (Assumptions)

#	Motive	Rationale
1	Financial Gain	Diamond necklaces are high-value items; personal theft is a common motive.
2	Insurance Fraud	Possible collusion with an insurer or internal policy manipulation.
3	Revenge or Personal Grievance	Employee dissatisfaction or conflict with management.
4	Cover-up of a Larger Scheme	Theft may be part of a broader internal theft operation.

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Run OnceRun Batch

query

*A diamond necklace is missing from a jewellery showroom. Three employees were prese

ClearExecute

7. Evidence That Should Be Checked

#	Evidence	How It Helps
1	CCTV footage (store interior, employee entrances, exit points)	Establishes who was where and when.
2	Employee access logs / badge swipes	Confirms entry/exit times.
3	Fingerprint/biometric data on the necklace or its packaging	Links suspect to the item.
4	Inventory/stock records	Confirms the necklace was recorded as present before the theft.
5	Employee statements & alibis	Identifies discrepancies or corroborations.
6	Security guard logs (if any)	Adds another layer of surveillance data.
7	Forensic evidence (e.g., shoe prints, tool marks)	May point to the method of removal.
8	Financial records of employees (salary, recent bonuses, debts)	May indicate motive.

8. Contradictions / Missing Information

#	Issue	Impact
1	No timestamps for key events	Limits ability to place suspects.
2	No CCTV footage or logs mentioned	Critical for confirming presence/absence.
3	No employee roles or responsibilities described	Hard to assess who could have handled the necklace.
4	No mention of security protocols (e.g., alarms, locks)	Unknown risk level.
5	No alibis provided	Cannot rule out any suspect.

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Dify

ADVANTAGES AND LIMITATIONS

Advantages

- Uses RAG for knowledge-grounded responses
- Quickly retrieves relevant information
- Reduces manual searching
- Provides natural-language answers
- Easy to interact with

Limitations

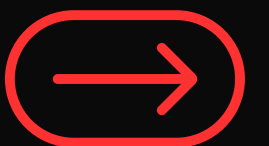
- Depends on the available knowledge base
- Incorrect or incomplete knowledge may affect results
- Requires AI service availability



FUTURE ENHANCEMENT

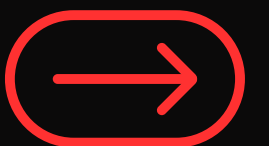
Future Enhancements

- Add more investigation cases to the knowledge base.
- Support multiple languages.
- Add image-based clue analysis.
- Add evidence comparison.
- Add suspect relationship mapping.
- Improve retrieval accuracy.
- Add voice-based investigation queries.



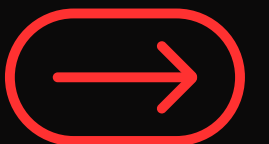
CONCLUSION

Mystery Investigation AI demonstrates how Generative AI + RAG can assist in investigation-related queries. The system retrieves relevant information from the knowledge base and provides it to the AI model for analysis. This approach helps users obtain clear, context-based and useful investigation responses instead of relying only on general AI knowledge.



REFERENCES

- Dify AI Platform Documentation
- Retrieval-Augmented Generation (RAG) Concepts
- Knowledge Retrieval Documentation
- Large Language Model (LLM) Documentation
- Investigation Knowledge Base Documents



THANK YOU

